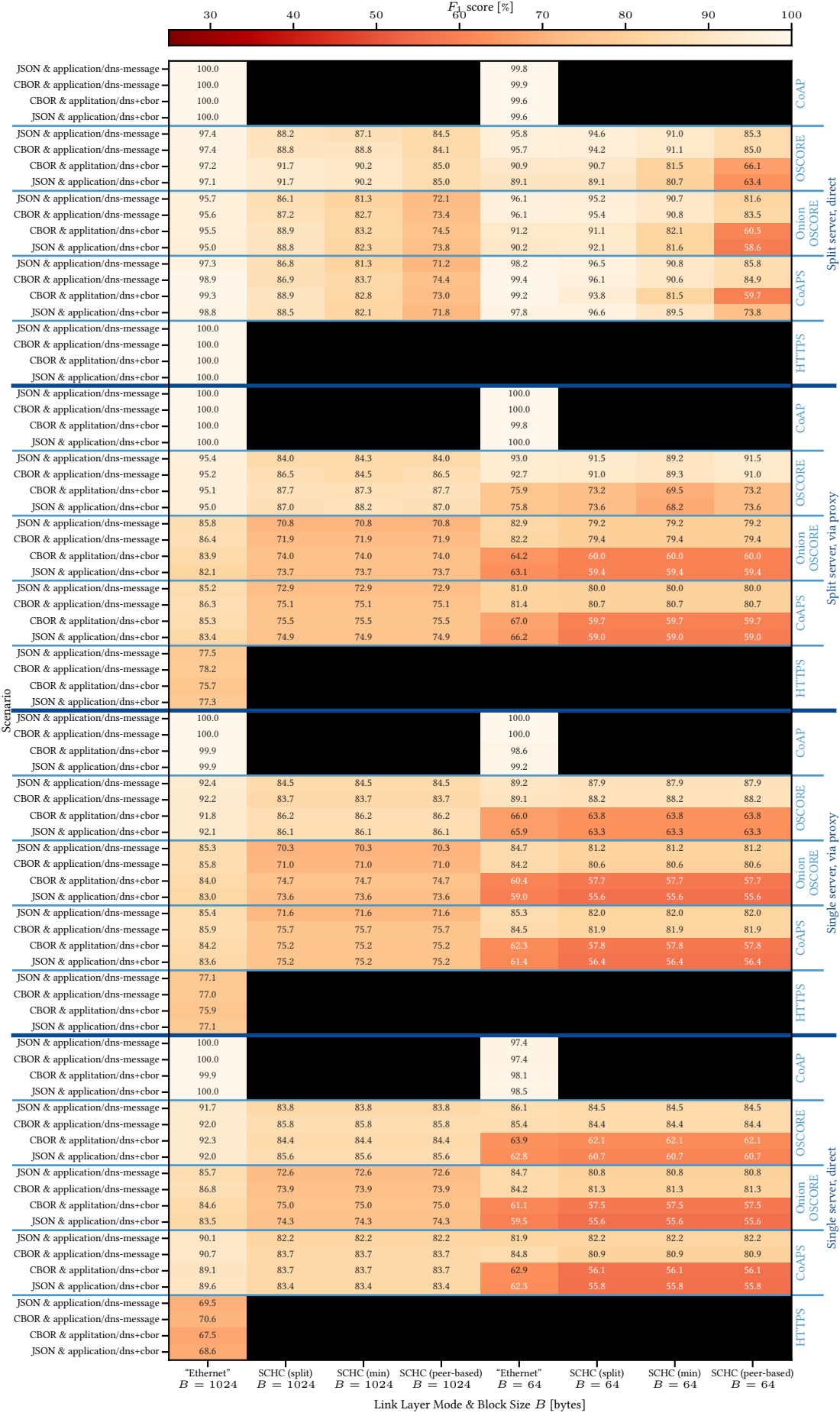
F_1 score [%]
30
40
50
60
70
80
90
100
Scenario
Link Layer Mode & Block Size B [bytes]
"Ethernet" $B = 1024$
SCHC (split) $B = 1024$
SCHC (min) $B = 1024$
SCHC (peer-based) $B = 1024$
"Ethernet" $B = 64$
SCHC (split) $B = 64$
SCHC (min) $B = 64$
SCHC (peer-based) $B = 64$
Split server, direct
JSON & application/dns-message 100.0 99.8
CBOR & application/dns-message 100.0 99.9
CBOR & applitation/dns+cbor 100.0 99.6
JSON & application/dns+cbor 100.0 99.6
CoAP
JSON & application/dns-message 97.4 88.2 87.1 84.5 95.8 94.6 91.0 85.3
CBOR & application/dns-message 97.4 88.8 88.8 84.1 95.7 94.2 91.1 85.0
CBOR & applitation/dns+cbor 97.2 91.7 90.2 85.0 90.9 90.7 81.5 66.1
JSON & application/dns+cbor 97.1 91.7 90.2 85.0 89.1 89.1 80.7 63.4
OSCORE
JSON & application/dns-message 95.7 86.1 81.3 72.1 96.1 95.2 90.7 81.6
CBOR & application/dns-message 95.6 87.2 82.7 73.4 96.1 95.4 90.8 83.5
CBOR & applitation/dns+cbor 95.5 88.9 83.2 74.5 91.2 91.1 82.1 60.5
JSON & application/dns+cbor 95.0 88.8 82.3 73.8 90.2 92.1 81.6 58.6
Onion OSCORE
JSON & application/dns-message 97.3 86.8 81.3 71.2 98.2 96.5 90.8 85.8
CBOR & application/dns-message 98.9 86.9 83.7 74.4 99.4 96.1 90.6 84.9
CBOR & applitation/dns+cbor 99.3 88.9 82.8 73.0 99.2 93.8 81.5 59.7
JSON & application/dns+cbor 98.8 88.5 82.1 71.8 97.8 96.6 89.5 73.8
CoAPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 100.0
HTTPS
Split server, via proxy
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 99.8
JSON & application/dns+cbor 100.0 100.0
CoAP
JSON & application/dns-message 95.4 84.0 84.3 84.0 93.0 91.5 89.2 91.5
CBOR & application/dns-message 95.2 86.5 84.5 86.5 92.7 91.0 89.3 91.0
CBOR & applitation/dns+cbor 95.1 87.7 87.3 87.7 75.9 73.2 69.5 73.2
JSON & application/dns+cbor 95.0 87.0 88.2 87.0 75.8 73.6 68.2 73.6
OSCORE
JSON & application/dns-message 85.8 70.8 70.8 70.8 82.9 79.2 79.2 79.2
CBOR & application/dns-message 86.4 71.9 71.9 71.9 82.2 79.4 79.4 79.4
CBOR & applitation/dns+cbor 83.9 74.0 74.0 74.0 64.2 60.0 60.0 60.0
JSON & application/dns+cbor 82.1 73.7 73.7 73.7 63.1 59.4 59.4 59.4
Onion OSCORE
JSON & application/dns-message 85.2 72.9 72.9 72.9 81.0 80.0 80.0 80.0
CBOR & application/dns-message 86.3 75.1 75.1 75.1 81.4 80.7 80.7 80.7
CBOR & applitation/dns+cbor 85.3 75.5 75.5 75.5 67.0 59.7 59.7 59.7
JSON & application/dns+cbor 83.4 74.9 74.9 74.9 66.2 59.0 59.0 59.0
CoAPS
JSON & application/dns-message 77.5
CBOR & application/dns-message 78.2
CBOR & applitation/dns+cbor 75.7
JSON & application/dns+cbor 77.3
HTTPS
Single server, via proxy
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 99.9 98.6
JSON & application/dns+cbor 99.9 99.2
CoAP
JSON & application/dns-message 92.4 84.5 84.5 84.5 89.2 87.9 87.9 87.9
CBOR & application/dns-message 92.2 83.7 83.7 83.7 89.1 88.2 88.2 88.2
CBOR & applitation/dns+cbor 91.8 86.2 86.2 86.2 66.0 63.8 63.8 63.8
JSON & application/dns+cbor 92.1 86.1 86.1 86.1 65.9 63.3 63.3 63.3
OSCORE
JSON & application/dns-message 85.3 70.3 70.3 70.3 84.7 81.2 81.2 81.2
CBOR & application/dns-message 85.8 71.0 71.0 71.0 84.2 80.6 80.6 80.6
CBOR & applitation/dns+cbor 84.0 74.7 74.7 74.7 60.4 57.7 57.7 57.7
JSON & application/dns+cbor 83.0 73.6 73.6 73.6 59.0 55.6 55.6 55.6
Onion OSCORE
JSON & application/dns-message 85.4 71.6 71.6 71.6 85.3 82.0 82.0 82.0
CBOR & application/dns-message 85.9 75.7 75.7 75.7 84.5 81.9 81.9 81.9
CBOR & applitation/dns+cbor 84.2 75.2 75.2 75.2 62.3 57.8 57.8 57.8
JSON & application/dns+cbor 83.6 75.2 75.2 75.2 61.4 56.4 56.4 56.4
CoAPS
JSON & application/dns-message 77.1
CBOR & application/dns-message 77.0
CBOR & applitation/dns+cbor 75.9
JSON & application/dns+cbor 77.1
HTTPS
Single server, direct
JSON & application/dns-message 100.0 97.4
CBOR & application/dns-message 100.0 97.4
CBOR & applitation/dns+cbor 99.9 98.1
JSON & application/dns+cbor 100.0 98.5
CoAP
JSON & application/dns-message 91.7 83.8 83.8 83.8 86.1 84.5 84.5 84.5
CBOR & application/dns-message 92.0 85.8 85.8 85.8 85.4 84.4 84.4 84.4
CBOR & applitation/dns+cbor 92.3 84.4 84.4 84.4 63.9 62.1 62.1 62.1
JSON & application/dns+cbor 92.0 85.6 85.6 85.6 62.8 60.7 60.7 60.7
OSCORE
JSON & application/dns-message 85.7 72.6 72.6 72.6 84.7 80.8 80.8 80.8
CBOR & application/dns-message 86.8 73.9 73.9 73.9 84.2 81.3 81.3 81.3
CBOR & applitation/dns+cbor 84.6 75.0 75.0 75.0 61.1 57.5 57.5 57.5
JSON & application/dns+cbor 83.5 74.3 74.3 74.3 59.5 55.6 55.6 55.6
Onion OSCORE
JSON & application/dns-message 90.1 82.2 82.2 82.2 81.9 82.2 82.2 82.2
CBOR & application/dns-message 90.7 83.7 83.7 83.7 84.8 80.9 80.9 80.9
CBOR & applitation/dns+cbor 89.1 83.7 83.7 83.7 62.9 56.1 56.1 56.1
JSON & application/dns+cbor 89.6 83.4 83.4 83.4 62.3 55.8 55.8 55.8
CoAPS
JSON & application/dns-message 69.5
CBOR & application/dns-message 70.6
CBOR & applitation/dns+cbor 67.5
JSON & application/dns+cbor 68.6
HTTPS